

Wave Casting for Hybrid-Scene Interactive Holography

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S1 MORE DETAILS OF WAVECAST

S1.1 Hologram Formulation

The goal of computer-generated holography is to synthesize a complex-valued wavefield on the SLM plane whose free-space propagation reconstructs the desired 3D scene. In WaveCast, the scene contribution to the hologram is expressed through a set of transport-aware interactions collected by wave casting,

$$\mathcal{I} = \mathcal{I}^v \cup \mathcal{I}^s, \quad (\text{S1})$$

where $\mathcal{I}^v$ and $\mathcal{I}^s$ denote volumetric and surface interactions, respectively. For each interaction $j \in \mathcal{I}$, we denote its projected SLM center as $\mathbf{u}_j$, its hologram-domain depth as z_j , and its channel-wise complex amplitude as a_c^j , which absorbs appearance, transmittance, and interface throughput.

Let $\mathbf{u}$ denote a pixel coordinate on the SLM plane and $\mathbf{x}(\mathbf{u})$ its physical 3D position. An interaction point $\mathbf{p}_j$ contributes a spherical wave to every SLM pixel:

$$U_c^j(\mathbf{u}) = \frac{a_c^j}{r_j(\mathbf{u})} \exp\left[i\left(k_c r_j(\mathbf{u}) + \phi_c^j\right)\right], \quad k_c = \frac{2\pi}{\lambda_c}. \quad (\text{S2})$$

where

$$r_j(\mathbf{u}) = \|\mathbf{x}(\mathbf{u}) - \mathbf{p}_j\| \quad (\text{S3})$$

is the propagation distance, λ_c is the wavelength of color channel c and ϕ_c^j is the initial phase assigned to interaction j . When the lateral displacement from the projected center $\mathbf{u}_j$ is small relative to the axial distance z_j , the distance can be approximated by the Fresnel expansion,

$$r_j(\mathbf{u}) \approx |z_j| + \frac{p^2 \|\mathbf{u} - \mathbf{u}_j\|^2}{2|z_j|}, \quad (\text{S4})$$

where p is the SLM pixel pitch. Inspired by the phase-initialization strategy of Maimone et al. [Maimone et al. 2017], we choose the object phase to compensate for the depth-dependent constant phase at the subhologram center. Specifically, we assign an initial phase as $\phi_c^j = -k_c |z_j|$. And we also absorb slowly varying scale factors into the complex amplitude a_c^j . The remaining spatially varying phase over the subhologram is therefore the quadratic Fresnel term, given by

$$\tilde{H}_c^j(\mathbf{u}) = a_c^j \exp\left[i \frac{\pi p^2 \|\mathbf{u} - \mathbf{u}_j\|^2}{\lambda_c z_j}\right]. \quad (\text{S5})$$

Diffraction-limited subhologram support. Ideally, a continuous spherical wave has nonzero support over the entire hologram plane. However, a pixelated SLM cannot represent arbitrary propagation

angles: its pixel pitch p imposes a maximum diffraction angle,

$$\theta_{\max,c} = \sin^{-1}\left(\frac{\lambda_c}{2p}\right). \quad (\text{S6})$$

Thus, an interaction at depth z_j contributes only within a finite footprint on the SLM. In pixel units, the diffraction-limited radius is

$$r_{\lambda_c}(z_j) = \frac{|z_j| \tan \theta_{\max,c}}{p} \approx \frac{\lambda_c |z_j|}{2p^2}. \quad (\text{S7})$$

We refer to this finite footprint as the interaction’s *subhologram*. Using a finite-support window W_T , implemented as a Tukey window in WaveCast, the subhologram contribution of interaction j is

$$H_c^j(\mathbf{u}) = \frac{a_c^j}{\gamma_c(z_j)} W_T\left(\frac{\|\mathbf{u} - \mathbf{u}_j\|}{r_{\lambda_c}(z_j)}\right) \exp\left[i \frac{\pi p^2 \|\mathbf{u} - \mathbf{u}_j\|^2}{\lambda_c z_j}\right]. \quad (\text{S8})$$

Here $\gamma_c(z_j) = \max(\sqrt{\pi r_{\lambda_c}^2(z_j)}, 1)$ normalizes each point’s contribution by the square root of its subhologram area, with a one-pixel lower bound to avoid amplifying very small footprints.

The final hologram is obtained by coherent superposition of all interaction-induced subholograms:

$$\mathcal{H}_c(\mathbf{u}) = \sum_{j \in \mathcal{I}^v \cup \mathcal{I}^s} H_c^j(\mathbf{u}). \quad (\text{S9})$$

This source-centric form matches the ray-tracing implementation: each ray discovers its own volumetric and surface interactions, and each interaction scatters a finite-support complex contribution to the SLM. Notably, the free-space propagation describes diffraction between the hologram plane to the reconstruction volume, but it does not by itself determine the scene-dependent interaction set $\mathcal{I}$.

S1.2 Representative Hybrid Scene Instantiation

The main paper formulates WaveCast over a heterogeneous scene $\mathcal{P} = \mathcal{P}^v \cup \mathcal{P}^s$ (Sec. 3.1), where $\mathcal{P}^v$ collects volumetric primitives and $\mathcal{P}^s$ collects surface primitives. This subsection describes how the two families are instantiated in our experiments and records the appearance/transport quantities consumed by the per-hit interaction set $\mathcal{I} = \mathcal{I}^v \cup \mathcal{I}^s$.

Volumetric primitives $\mathcal{P}^v$. We instantiate $\mathcal{P}^v$ with anisotropic 3D Gaussians following 3D Gaussian Splatting [Kerbl et al. 2023]. Each Gaussian G_i is parameterized by a center $\xi_i \in \mathbb{R}^3$, a positive-definite covariance $\Sigma_i \in \mathbb{R}^{3 \times 3}$, a base opacity $\sigma_i \in [0, 1]$, and spherical-harmonic (SH) appearance coefficients $\mathbf{k}_i$, so that the view-dependent color along query direction $\mathbf{d}$ is $\mathbf{c}_i(\mathbf{d}) = \text{SH}_i(\mathbf{d})$. Its continuous density at any spatial point $\mathbf{p} \in \mathbb{R}^3$ is

$$G_i(\mathbf{p}) = \exp\left(-\frac{1}{2}d_i^2(\mathbf{p})\right), \quad d_i^2(\mathbf{p}) = (\mathbf{p} - \xi_i)^\top \Sigma_i^{-1}(\mathbf{p} - \xi_i), \quad (\text{S10})$$

where $d_i(\mathbf{p})$ is the Mahalanobis distance from $\mathbf{p}$ to the Gaussian center under Σ_i (i.e., a covariance-normalized radial distance that

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equals 1 on the one-standard-deviation isosurface of G_i). The effective opacity contributed by G_i at $\mathbf{p}$ is

$$\alpha_i(\mathbf{p}) = \sigma_i \exp\left(-\frac{1}{2}d_i^2(\mathbf{p})\right), \quad (\text{S11})$$

i.e., the stored opacity attenuated by the Gaussian’s spatial profile at the evaluation point. The complete volumetric set is $\mathcal{P}^v = \{(\xi_i, \Sigma_i, \sigma_i, \mathbf{k}_i)\}_{i=1}^{N_v}$.

Gaussian splatting rasterization for reference. The standard 3DGS pipeline [Kerbl et al. 2023] synthesizes an image by *splatting* the 3D Gaussians onto the image plane. Using the EWA approximation [Zwicker et al. 2001], the projected 2D covariance is

$$\Sigma'_i = \mathbf{J} \mathbf{M}_{\text{view}} \Sigma_i \mathbf{M}_{\text{view}}^\top \mathbf{J}^\top, \quad (\text{S12})$$

where $\mathbf{J}$ is the projective Jacobian and $\mathbf{M}_{\text{view}}$ the world-to-camera transform. The screen-space footprint at image-plane coordinate $\mathbf{s} \in \mathbb{R}^2$ is

$$g_i(\mathbf{s}) = \exp\left(-\frac{1}{2}(\mathbf{s} - \xi'_i)^\top (\Sigma'_i)^{-1} (\mathbf{s} - \xi'_i)\right), \quad (\text{S13})$$

with ξ'_i the projection of ξ_i . The pixel color is then assembled by front-to-back alpha blending,

$$\mathbf{C}_{\text{out}}(\mathbf{s}) = \sum_{i \in \mathcal{N}_s} \mathbf{c}_i(\mathbf{d}) \alpha_i(\mathbf{s}) \prod_{j < i} (1 - \alpha_j(\mathbf{s})), \quad \alpha_i(\mathbf{s}) = \sigma_i g_i(\mathbf{s}), \quad (\text{S14})$$

where $\mathcal{N}_s$ is the depth-sorted subset of Gaussians overlapping pixel $\mathbf{s}$. A key property of Equation (S14) is that the rendered pixel is a *collective* contribution of all Gaussians along the viewing ray, combined through transmittance-weighted alpha compositing. In most cases, no individual Gaussian carries an independent rendering meaning of its own; only the cumulative product $\prod (1 - \alpha_j)$ and the weighted sum determine what is observed at $\mathbf{s}$. This collective property is what enables the segment-level virtual-source aggregation we exploit in Sec. 3.2 of the main paper: Gaussians along the same ray segment can be summarized by aggregated transport moments without greatly changing the segment’s observed appearance, opening the door to substantial wave-domain acceleration.

Surface primitives $\mathcal{P}^s$. We instantiate $\mathcal{P}^s$ with triangle meshes, which are given by

$$\mathcal{P}^s = \{\mathcal{M}_k\}_{k=1}^{N_s}, \quad \mathcal{M}_k = (V_k, F_k, \Psi_k), \quad (\text{S15})$$

where $V_k = \{\mathbf{v}_{k,\ell} \in \mathbb{R}^3\}$ are vertices, F_k are triangular faces, and Ψ_k collects material attributes (surface normal, albedo, refractive index, interface type). Meshes act as *surface interaction primitives*: they bound each ray segment, induce hard occlusion, and dispatch reflection or refraction via $\text{INTERFACE RESPONSE}(\cdot, \Psi_k)$ as in Algorithm S1. A mesh hit thus contributes both a surface entry to $\mathcal{I}^s$ and, when non-diffuse, a secondary segment to subsequent traversal.

S1.3 Algorithmic Summary of Wave Casting

In Algorithm S1, we provide an algorithmic summary of WaveCast’s ray-wave hologram rendering procedure, showing how wave casting collects surface and volumetric interactions and scatters their diffraction-limited subhologram contributions to the SLM. Given a heterogeneous scene, a target view, and target holographic display

parameters, the algorithm traces each target-view ray through volumetric and surface primitives, records transport-aware interactions, and scatters their diffraction-limited subhologram contributions to the complex SLM buffer. Gaussian hits are processed in front-to-back order with transmittance-aware compositing weights, whereas mesh hits introduce discrete surface events that either terminate the path or generate reflected/refracted secondary rays. For each recorded interaction, we compute an effective image depth, map it to the target reconstruction volume, and evaluate the Fresnel contribution within the corresponding subhologram support.

This algorithm corresponds to Mode 1: per-hit scatter, which forms the conceptual foundation of WaveCast. It directly realizes the core ray-wave idea: transport-aware surface and volumetric interactions are discovered by wave casting and converted into diffraction-limited subhologram contributions. Mode 2 and Mode 3 preserve this principle while introducing segment-level abstraction and custom tile-based gather-style CUDA kernel for efficient execution.

S1.4 Recursive Effective Depth Computation

Conventional ray- and path-tracing pipelines are designed primarily to estimate pixel radiance. Although they explicitly trace reflection and refraction paths and identify the geometric interactions contributing to each pixel, their standard depth outputs typically record either the camera-space depth of a geometric hit or the depth of the first visible surface. Neither quantity represents the apparent image depth induced by the preceding optical interfaces. This distinction is largely irrelevant for conventional appearance rendering, but is important for hologram synthesis, where the assigned depth directly determines the reconstruction focus.

For interactions reached after reflection or refraction, we therefore use the apparent image depth rather than the raw geometric hit depth for subhologram support and Fresnel phase calculations. We compute this depth by recursively applying standard paraxial Gaussian imaging principles [Born and Wolf 1999; Hecht 2017] to the preceding optical interfaces along the traced path. The resulting *effective image depth* $\tilde{z}_j$ captures the dominant axial image shift required for holographic focus.

For example, a point seen in a planar mirror should reconstruct at its reflected image position behind the mirror; a point viewed through a planar refractive interface appears at a compressed axial depth; and a point reflected by a spherical mirror may form an image at a finite distance determined by the mirror equation. This subsection details the routine $\text{EFFECTIVEIMAGEDEPTH}(\cdot, \mathcal{B})$ used in Algorithm S1. The resulting $\tilde{z}_j$ is mapped into the display’s target reconstruction range by $\text{MAPTOTARGETDEPTH}(\cdot)$ before being used as z_j in Equations (S7) and (S8).

Interface stack. For each ray, we maintain an ordered interface stack

$$\mathcal{B} = [\beta_1, \beta_2, \dots, \beta_Q], \quad \beta_q = (\Psi_q^s, \mathbf{p}_q^s, \eta_q^s), \quad (\text{S16})$$

matching the entries pushed in Algorithm S1: Ψ_q^s stores the interface type, surface normal, refractive index, and local signed curvature; $\mathbf{p}_q^s$ is the corresponding surface hit position; and η_q^s is the Fresnel throughput already used to update T_{in} . The throughput η_q^s is not

consumed by the depth recursion and is retained only for the transmittance bookkeeping. For the depth recursion, we read out of Ψ_q^s the interface category $\chi_q \in \{\text{mirror, refraction}\}$, the relative refractive index $v_q = n_q^{\text{out}}/n_q^{\text{in}}$ for refraction, and the signed radius of curvature R_q (equivalently, the signed mean curvature $\kappa_q = 1/R_q$). We adopt the convention that $R_q > 0$ when the center of curvature lies on the camera side of the interface, so that concave-toward-camera mirrors and refractive surfaces have positive R_q . Planar interfaces correspond to $R_q = \infty$ and $\kappa_q = 0$.

Let $\mathbf{p}_0^s$ be the camera origin and $\mathbf{p}_{Q+1}^s$ be the final interaction position. We define the segment length along the broken ray as

$$d_q = \|\mathbf{p}_{q+1}^s - \mathbf{p}_q^s\|, \quad q = 0, 1, \dots, Q. \quad (\text{S17})$$

Per-interface paraxial transform. We approximate each interface by its local paraxial imaging behavior along the traced ray. Throughout the recursion, s and s' denote *unfolded* apparent distances along the broken ray: s is measured from interface β_q outward to the next vertex $\mathbf{p}_{q+1}^s$, and s' is the apparent distance from β_q back along the camera-side path at which the image of that vertex appears. Both are taken positive. Under our sign convention, the standard on-axis imaging [Born and Wolf 1999; Hecht 2017] take the form:

$$\frac{1}{s} - \frac{1}{s'} = \frac{2}{R_q} \quad (\text{spherical mirror}), \quad (\text{S18})$$

$$\frac{n_q^{\text{out}}}{s} + \frac{n_q^{\text{in}}}{s'} = \frac{n_q^{\text{out}} - n_q^{\text{in}}}{R_q} \quad (\text{spherical refractive interface}), \quad (\text{S19})$$

where the index labels in Equation (S19) reflect that, in our wave-casting setup, light is traced camera-outward so the actual scene point lies in medium n_q^{out} and the apparent image is sought in medium n_q^{in} . Solving for s' gives the per-interface transform

$$\mathcal{T}_{\beta_q}(s) = s' = \begin{cases} \frac{sR_q}{R_q - 2s}, & \chi_q = \text{mirror}, \\ \frac{R_q s}{(v_q - 1)s + v_q R_q}, & \chi_q = \text{refraction}. \end{cases} \quad (\text{S20})$$

In the planar limit $R_q \rightarrow \infty$, these reduce to $\mathcal{T}_{\beta_q}(s) = s$ for a planar mirror (image at the unfolded distance) and $\mathcal{T}_{\beta_q}(s) = s/v_q$ for a planar refractive interface (apparent depth compressed by the relative refractive index).

Recursive composition. Starting from the final interaction, we walk backward through the interface stack and recursively accumulate the apparent distance:

$$\begin{aligned} \tilde{s}_Q &= d_Q, \\ \tilde{s}_{q-1} &= d_{q-1} + \mathcal{T}_{\beta_q}(\tilde{s}_q), \quad q = Q, Q-1, \dots, 1. \end{aligned} \quad (\text{S21})$$

Each step composes free-space propagation over d_{q-1} with the paraxial image formation of interface β_q . The effective image depth is the final apparent distance from the camera,

$$\tilde{z}_j = \tilde{s}_0 = \text{EFFECTIVEIMAGEDEPTH}(\mathbf{p}_{Q+1}^s, \mathcal{B}). \quad (\text{S22})$$

For volumetric hits inside a secondary segment, the same stack $\mathcal{B}$ is applied to each hit before segment-level aggregation, so reflected and refracted Gaussian content is assigned an image-space reconstruction depth.

Special cases and model scope. If $\mathcal{B} = \emptyset$, then $\tilde{z}_j = d_0$, matching the direct camera-to-hit distance. For a planar mirror at a distance d_0 from the camera, $\tilde{z}_j = d_0 + d_1$ corresponds to the unfolded mirror path. For a spherical mirror, Equation (S18) gives the familiar finite image distance as a function of object distance and signed curvature. For a planar refractive interface with a relative index v_1 , $\tilde{z}_j = d_0 + d_1/v_1$ recovers the standard apparent-depth compression. Our computation currently rely on a paraxial model and primarily captures the dominant axial image shift required for holographic focus. Higher-order off-axis aberrations such as coma and astigmatism are currently not considered.

Visualizing recursive depth composition. To make the effect of recursive depth composition directly visible, we construct a qualitative stress-test scene in which three planar mirrors surround an object and generate a sequence of repeated, multi-bounce reflections. This infinite-mirror configuration deliberately amplifies the differences between alternative depth assignments, which may be subtle under a single reflection. As shown in Figure S1, raw geometric depth records the physical camera-space locations of the ray-traced interactions, while the interface-surface depth assigns reflected content to the corresponding mirror surfaces. Neither preserves the progressive apparent-depth ordering produced by repeated reflections. Our effective-depth formulation instead recursively composes the preceding optical interfaces and assigns successive mirror images to progressively deeper image planes.

S1.5 Tukey-Windowed subhologram Accumulation

The diffraction-limited radius $r_{\lambda_c}(z_j)$ in Equation (S7) defines the finite support of each subhologram, centered at the SLM position $\mathbf{u}_j$. A hard binary cutoff, however, can introduce visible color banding artifacts. Small depth variations lead to discrete changes in the rounded subhologram radius, causing neighboring interactions to abruptly cover different sets of SLM pixels. To suppress boundary ringing while preserving finite support, we use a radial Tukey window, a standard cosine-tapered window commonly used in spectral analysis [Harris 1978]. The Tukey window smooths this footprint transition near the boundary, reducing the abrupt pixel-level changes while preserving the same compact support. Let

$$\ell_j(\mathbf{u}) = \frac{\|\mathbf{u} - \mathbf{u}_j\|}{r_{\lambda_c}(z_j)} \quad (\text{S23})$$

be the normalized radial coordinate within the subhologram footprint. The Tukey window with roll-off parameter $\tau \in [0, 1]$ is defined as

$$W_T(\ell_j; \tau) = \begin{cases} 1, & 0 \leq \ell_j \leq 1 - \tau, \\ \frac{1}{2} \left[1 + \cos \left(\pi \frac{\ell_j - (1 - \tau)}{\tau} \right) \right], & 1 - \tau < \ell_j \leq 1, \\ 0, & \ell_j > 1. \end{cases} \quad (\text{S24})$$

Here $\tau \rightarrow 0$ recovers the hard-support case, while larger τ increases the cosine roll-off near the footprint boundary. In Equation (S8), we use $W_T(\ell_j(\mathbf{u}); \tau)$ to apodize each interaction's subhologram without changing its finite support. In all experiments, we use $\tau = 0.4$, which lies in the stable high-quality regime across metrics and execution modes. Figure S4 shows the effect of sweeping τ : moderate roll-off

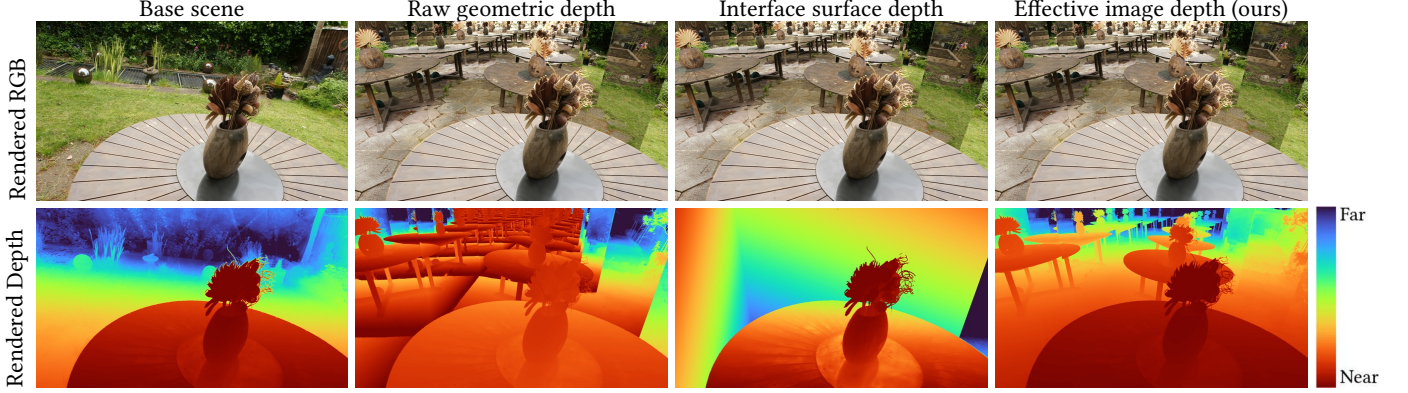


Fig. S1. **Comparison of depth assignments under recursive reflections.** We deliberately arrange three planar mirrors around a central object to produce repeated multi-bounce reflections and amplify the differences between depth assignments. The first column shows the base scene without the mirror arrangement. The remaining columns show the same hybrid-scene RGB rendering using raw geometric, interface-surface, and effective image-depth assignments, respectively. Raw geometric depth uses the camera-space depth of the ray-traced interaction without accounting for its reflected image position; interface-surface depth assigns reflected content to the corresponding mirror surface; and our effective image depth recursively places each reflection at the apparent image position induced by the preceding interfaces. Consequently, the repeated mirror images form progressively deeper layers only in the effective-depth map. The RGB rendering is identical across the three hybrid columns because the depth mode changes only the depth supplied to hologram synthesis. Each depth map is normalized independently for visualization.

values improve reconstruction quality over the hard-support case and suppress ringing artifacts, while overly large roll-off values reduce the effective flat region of each subhologram.

S2 ADDITIONAL IMPLEMENTATION DETAILS

S2.1 WaveCast Framework

WaveCast’s wave-casting stage is implemented on top of NVIDIA OptiX 7 [Parker et al. 2010], building upon the 3DGRT implementation [Moenne-Loccoz et al. 2024]. We use a single OptiX pipeline to handle mesh traversal, Gaussian traversal, surface-event dispatch, and per-hit transport bookkeeping. The SLM-side Fresnel synthesis is executed either inline as scatter accumulation for Mode 1 and Mode 2, or by a separate tile-based CUDA gather kernel for Mode 3.

OptiX programs. The pipeline uses the standard OptiX program types. The raygen program runs the outer wave-casting loop in Algorithm S1: it alternates between mesh and Gaussian tracing, maintains the transmittance T and interface stack $\mathcal{B}$, and either scatters Fresnel contributions directly or buffers emitters for the gather kernel. The closest-hit program handles surface events by dispatching according to the material stored in Ψ^s , such as diffuse, mirror, or glass. For reflective or refractive surfaces, it computes the redirected ray and pushes the corresponding interface record $\beta_q = (\Psi_q^s, \mathbf{p}_q^s, \eta_q^s)$ onto $\mathcal{B}$. The any-hit program maintains a small front-to-back k -buffer for Gaussian intersections, allowing the ray-generation program to consume volumetric hits in depth order and update transmittance. The miss program marks the ray as escaped and terminates the current segment.

Acceleration structures. We maintain separate acceleration structures for surface and volumetric primitives. Triangle meshes are stored in a single OptiX geometry acceleration structure (GAS) over all loaded mesh assets. Static meshes are built once at scene load, while deformable meshes are refit per frame and rebuilt only when

topology changes. Gaussian primitives are enclosed by tight proxy geometry, implemented as icosahedra by default and sized according to their 3σ covariance extents. These proxies form a separate GAS for volumetric traversal.

Tile-based gather CUDA kernel. Mode 3 realizes the tiled gather formulation with a four-stage CUDA pipeline. We partition the SLM into fixed-size tiles and first count how many tiles each emitter overlaps. A prefix scan allocates slots in a flat pair buffer, after which we generate one (tileId, emitterIdx) pair for each overlapped tile. The pairs are then sorted by tileId, and a range lookup is built so that each tile can access a contiguous list of emitters.

In the gather stage, one CUDA block processes one non-empty tile. The block streams the tile’s emitter list through shared memory in fixed-size chunks. Each thread owns one output pixel, evaluates Equation (S8) for that pixel against the shared emitters, accumulates the complex field in registers, and finally writes the result to global memory with a single non-atomic store. In our implementation, we use 16×16 tiles, so each CUDA block contains 256 threads and each staged emitter can be reused by all pixels in the tile. This realizes the three acceleration mechanisms described in the main paper: non-atomic output writes, independent tile-level CUDA parallelism, and shared-memory reuse with register-resident accumulation.

S2.2 Hardware setup

Prototype. Our holographic display prototype uses a FISBA READY-Beam IND fiber-coupled RGB module (638, 520, 450 nm). The illumination is collimated, polarized along the SLM’s active axis, and incident on a HOLOEYE PLUTO-2.1 (VIS-014) LCOS phase-only SLM (1920×1080 , $8 \mu\text{m}$ pitch, 8-bit). The modulated wavefront passes a 4- f relay ($f_1 = 150\text{mm}$, $f_2 = 75\text{mm}$) with a spatial filter at the Fourier plane that suppresses higher diffraction orders, and is recorded by a ZWO ASI533MC Pro color CMOS sensor on a motorized linear translation stage. The 4 f magnification is chosen as

Algorithm S1 WaveCast: Ray-Wave Hologram Rendering over Heterogeneous-Primitive Scene**Require:** Heterogeneous scene $\mathcal{P} = \mathcal{P}^v \cup \mathcal{P}^s$; target view; holographic display parameters, including SLM pixel pitch p , wavelengths $\{\lambda_c\}$ **Ensure:** Complex hologram fields $\{\mathcal{H}_c(\mathbf{u})\}$ on the SLM plane

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1:  $\mathcal{H}_c(\mathbf{u}) \leftarrow 0 \ \forall c, \mathbf{u}$  ▷ complex SLM buffers
2: for all target-view samples  $(u, v)$  in parallel do
3:    $(\mathbf{o}_0, \mathbf{d}_0) \leftarrow \text{GENERATECAMERARAY}(u, v)$ 
4:    $T_{\text{in}} \leftarrow 1; \ \mathcal{B} \leftarrow \emptyset$  ▷ path transmittance and interface stack
5:   for  $j = 1, \dots, N_{\text{bounce}}$  do ▷ ray segment / bounce index
6:     Query mesh BVH for nearest surface event:  $(t^s, \mathbf{p}^s, \Psi^s)$ 
7:     Traverse  $\mathcal{P}^v$  over  $[0, t^s]$  and collect Gaussian hits sorted by depth:  $\{(t_i^v, \mathbf{p}_i^v, \alpha_i, \mathbf{c}_i)\}_{i=1}^N$ 
8:      $T \leftarrow T_{\text{in}}$ 
9:     for  $i = 1, \dots, N$  do ▷ volumetric interactions in segment  $\mathcal{S}_j$ 
10:       $\mathbf{c}_i \leftarrow \text{EVALUATEAPPEARANCE}_i(\mathbf{d}_{j-1})$  ▷ e.g., SH view-dependent color
11:       $w_i \leftarrow T \alpha_i$  ▷ transport-aware compositing weight
12:       $\tilde{z}_i \leftarrow \text{EFFECTIVEIMAGEDEPTH}(t_i^v, \mathcal{B})$ 
13:       $z_i \leftarrow \text{MAPTOTARGETDEPTH}(\tilde{z}_i)$ 
14:       $\text{SCATTERSUBHOLOGRAM}(\mathbf{p}_i^v, z_i, w_i \mathbf{c}_i)$ 
15:       $T \leftarrow T(1 - \alpha_i)$  ▷ front-to-back transmittance update
16:   end for
17:   if  $\text{HITSURFACE}(t^s)$  then
18:      $\mathbf{c}^s \leftarrow \text{EVALUATESURFACERESPONSE}(\mathbf{p}^s, \Psi^s, \mathbf{d}_{j-1})$ 
19:      $\tilde{z}^s \leftarrow \text{EFFECTIVEIMAGEDEPTH}(\mathbf{p}^s, \mathcal{B})$ 
20:      $z^s \leftarrow \text{MAPTOTARGETDEPTH}(\tilde{z}^s)$ 
21:      $\text{SCATTERSUBHOLOGRAM}(\mathbf{p}^s, z^s, T \mathbf{c}^s)$  ▷ surface contribution weighted by remaining transmittance
22:     if  $\Psi^s$  is reflective or refractive then
23:        $(\mathbf{o}_j, \mathbf{d}_j, \eta^s) \leftarrow \text{INTERFACERESPONSE}(\mathbf{p}^s, \mathbf{d}_{j-1}, \Psi^s)$ 
24:        $T_{\text{in}} \leftarrow T \eta^s$  ▷ continue along secondary ray
25:        $\mathcal{B} \leftarrow \mathcal{B} \cup \{(\Psi^s, \mathbf{p}^s, \eta^s)\}$  ▷ update interface stack
26:     else
27:       break ▷ diffuse event terminates the path
28:     end if
29:   else
30:     break ▷ ray exits scene bound
31:   end if
32:   if  $T_{\text{in}} < \epsilon$  then
33:     break ▷ negligible remaining throughput
34:   end if
35: end for
36: end for
37: return  $\{\mathcal{H}_c(\mathbf{u})\}_{c \in \{R, G, B\}}$ 

38: procedure  $\text{SCATTERSUBHOLOGRAM}(\mathbf{p}_j, z_j, a^j)$ 
39:    $\mathbf{u}_j \leftarrow \text{PROJECTTOSLM}(\mathbf{p}_j)$  ▷ lateral center on the SLM
40:   for  $c \in \{R, G, B\}$  do
41:      $r_{\lambda_c}(z_j) \leftarrow \lambda_c |z_j| / (2p^2)$  ▷ diffraction-limited support
42:     for all  $\mathbf{u}$  with  $\|\mathbf{u} - \mathbf{u}_j\| \leq r_{\lambda_c}(z_j)$  do
43:        $\mathcal{H}_c(\mathbf{u}) \mathrel{+}= H_c^j(\mathbf{u}; a_c^j, z_j)$  ▷ Tukey-windowed Fresnel contribution
44:     end for
45:   end for
46: end procedure

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$f_2/f_1 = 0.5$ so that the full SLM area is imaged onto the camera sensor in its entirety, while the demagnified SLM-pixel pitch ($4.0 \mu\text{m}$) remains above the $3.76 \mu\text{m}$ camera pitch.

Calibration. Hardware imperfection causes the captured results to deviate from the simulated images. We address this with a learnable parameter in the complex SLM-plane source field to cover the correction inspired by the camera-in-the-loop calibration[Choi et al.

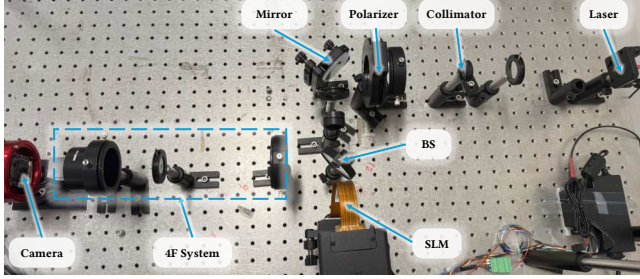
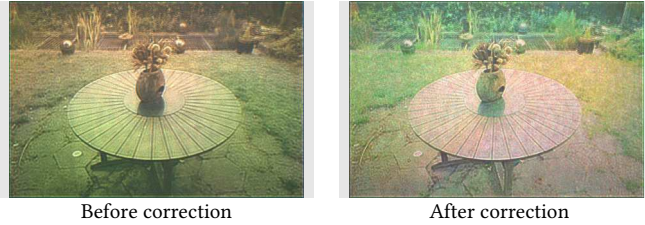


Fig. S2. Hardware Setup.

Fig. S3. **Effect of per-channel color recalibration.** (a) Color-multiplexed capture using shared camera settings. (b) Sequential RGB capture with channel-specific radiometric tuning and a per-pixel SLM phase LUT.

2022; Peng et al. 2020]. To be specific, we captured $N = 1300$ pairs of SLM phase maps spanning five depth planes (uniformly distributed from 0 mm to 30mm). The phases were generated with a stochastic gradient descent approach with randomized iterations and learning rates out of the ImageNet dataset [Deng et al. 2009]. And we trained a standalone SLM field to absorb the hardware imperfections. We applied the learned SLM plane with the complex field generated by our approaches, and then performed the phase encoding afterward.

Color synchronization. Our results are captured in a color sequential manner with synchronized SLM and laser updates within a single exposure. Slight synchronization offset due to the external trigger can cause a lag between the SLM phase and laser updates, leading to color bleeding and elevated artifacts in the captured results, see Figure S3(left). On the other hand, similar to prior work [Choi et al. 2025], when each color channel is acquired separately with wavelength-specific exposure and gain settings [Peng et al. 2020], the reconstructed colors improve substantially. Figure S3(right) shows holograms captured with per-channel radiometric calibration and a per-pixel SLM phase LUT for each color, before combining the measurements. This procedure substantially improved the brightness and color balance of the optical reconstruction.

S2.3 Additional Details for Evaluation

We compare against three CGH baselines provided by the public hsplat codebase¹: *Gaussian Wave Splatting (GWS)* [Choi et al. 2025], which synthesizes a hologram directly from a 3D Gaussian field by splatting per-Gaussian wavefront contributions to the SLM; *Mesh Adaptive Wavefront Blending (Mesh-AWB)*, which generates a hologram from an explicit triangle mesh by blending per-triangle wavefront primitives weighted by visibility and view-dependent shading;

and *Mesh Silhouette-based Occlusion (Mesh-Sil)*, a mesh-based variant that handles inter-surface occlusion through silhouette tests rather than wavefront-domain compositing.

For the GWS baseline, we use the official checkpoints released by the original authors so that the comparison reflects the recommended pretrained-model setting rather than any reimplementation on our side. The triangle meshes consumed by the mesh-based baselines are taken from Nerf2Mesh [Tang et al. 2023], which provides per-scene meshes extracted from neural radiance fields and is also the source used by the hsplat codebase. For WaveCast, we train the 3DGS scenes using 3DGRT [Moenne-Loccoz et al. 2024] so that the volumetric input is consistent with our wave-casting pipeline. To keep the comparison fair, we align two budgets across methods: (i) the training-view image resolution we use for 3DGRT matches the resolution at which hsplat / Nerf2Mesh were trained on the same scene, and (ii) when a scene is also evaluated under a mesh-based baseline, we cap the Nerf2Mesh extraction so that its triangle count is comparable to (and not below) the count used by the mesh-based baselines on the same scene. Both controls hold the underlying geometric capacity roughly fixed, isolating the comparison to the wave-synthesis stage.

Mesh asset preparation for mesh-based baselines. For mesh-based baseline evaluations, we convert the Nerf2Mesh reconstructed meshes [Tang et al. 2023] to the coordinate convention used by the hsplat-based pipeline. Nerf2Mesh meshes are extracted in the normalized NeRF training frame, whereas 3DGS-style frameworks, including hsplat, operate in the original COLMAP camera coordinate system. We therefore apply a similarity transform (R, t, s) to map each reconstructed mesh back to the COLMAP-aligned scene frame, correcting the axis convention, origin, and scale.

S3 ADDITIONAL RESULTS AND DISCUSSIONS

S3.1 Additional Results for System Analysis

S3.1.1 Effect of the Tukey-window roll-off parameter. As described in Section S1.5, the Tukey window is introduced to smooth the finite subhologram support and reduce artifacts from hard boundary truncation. We evaluate $\tau \in \{0.0, 0.2, 0.4, 0.6, 1.0\}$ across the three holographic rendering modes of our method. As shown in Figure S4, replacing the box window ($\tau = 0$) with a mild Tukey roll-off significantly improves reconstruction quality, increasing PSNR/SSIM and reducing LPIPS. The improvement is most pronounced from $\tau = 0$ to $\tau = 0.2$, after which the metrics enter a stable high-quality regime and vary only mildly across $\tau = 0.4$ – 1.0 .

Consistent with this trend, the visual results in Figure S5 show pronounced banding artifacts for $\tau = 0$, which are effectively removed once $\tau \geq 0.2$. To quantify the corresponding frequency-domain attenuation, we report the outer-band energy retention. The outer band is defined as the annular region spanning 60%–100% of the Nyquist radius, with its inner boundary indicated by the red dashed circle in Figure S5. We define the outer-band energy retention as the fraction of the angular-spectrum power within the Nyquist disk that lies in this outer band, normalized by the corresponding fraction for the box window ($\tau = 0$). As τ increases from 0 to 1.0, the retained outer-band energy decreases from 100% to 13.3%, indicating

¹<https://github.com/computational-imaging/hsplat>

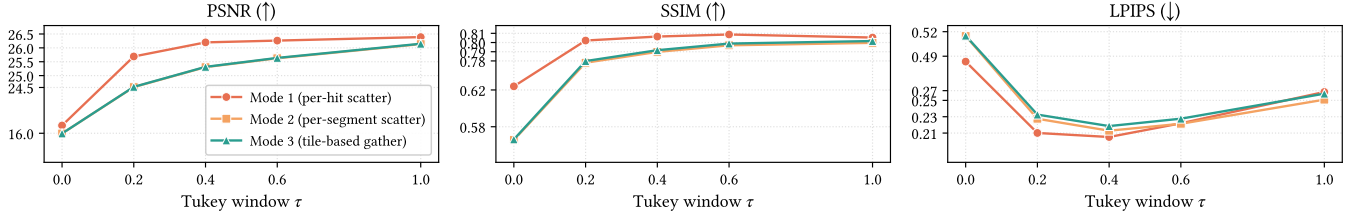


Fig. S4. **Effect of the Tukey-window roll-off parameter τ on reconstruction quality.** All-in-focus reconstruction quality is evaluated for $\tau \in \{0.0, 0.2, 0.4, 0.6, 1.0\}$ across the three holographic ray-tracing modes. The panels report PSNR ($\uparrow$), SSIM ($\uparrow$), and LPIPS ($\downarrow$), respectively, averaged over 70 scene-view AIF reconstructions. Quality improves sharply from the boxcar window ($\tau = 0$) to mild Tukey roll-offs ($\tau = 0.2$ – 0.4), after which it largely saturates across all modes. We use $\tau = 0.4$ as the default setting, as it lies in a stable high-quality regime across metrics and variants.

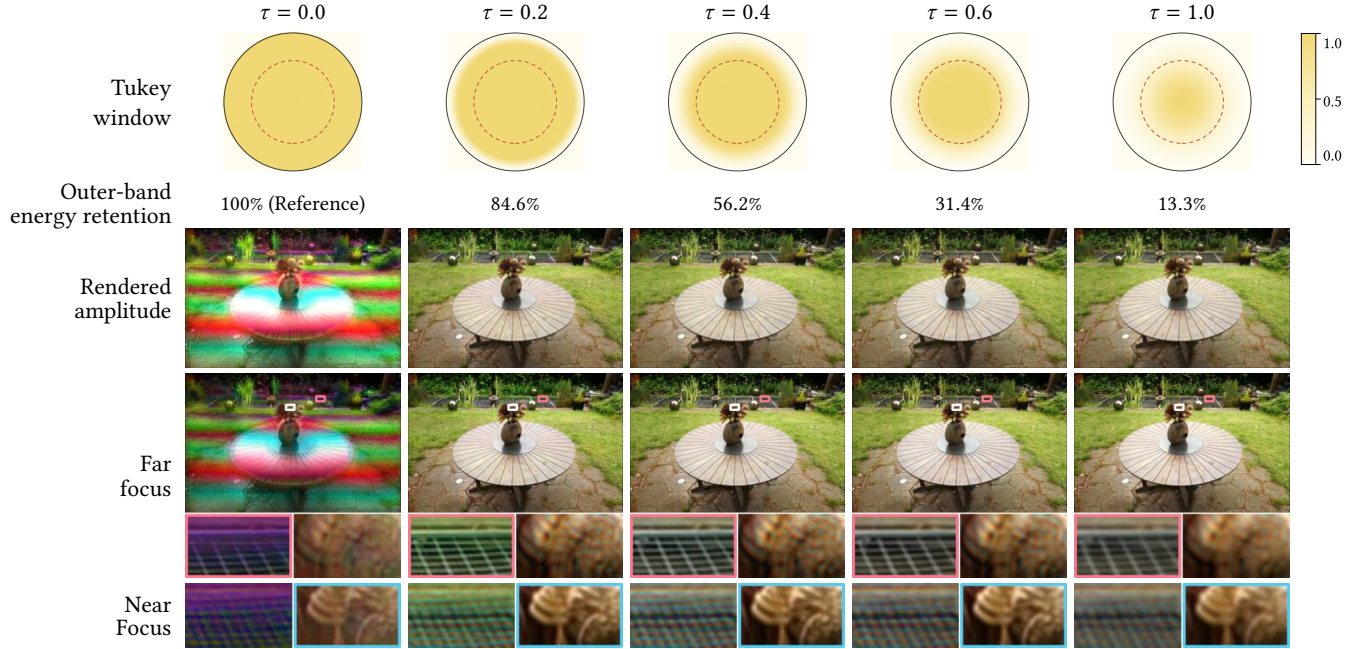


Fig. S5. **Effect of the Tukey-window roll-off parameter τ .** The first row shows the Tukey windows, whose radius spans the diffraction-angle range supported by the SLM. The red dashed circle marks 60% of the maximum diffraction angle and defines the inner boundary of the outer band. The value below each window is the fraction of the sub-hologram’s diffracted power in this outer band, normalized by the corresponding fraction for the box window ($\tau = 0$). The remaining rows show the rendered amplitudes and enlarged far- and near-focus regions. Increasing τ suppresses boundary-induced banding and color artifacts, while an excessive roll-off slightly reduces the sharpness of in-focus details. We use $\tau = 0.4$ as the default.

progressively stronger attenuation near the boundary of the frequency support. This attenuation suppresses the boundary-induced artifacts, particularly in the in-focus regions, where fine details and high-frequency fringes make these artifacts most visible. However, an excessively strong roll-off also removes useful high-frequency information, resulting in a slight reduction in the sharpness and local contrast of in-focus details. We therefore set $\tau = 0.4$ as the default: it retains 56.2% of the outer-band energy, effectively suppresses boundary-induced artifacts, and preserves sharp details in the in-focus regions (Figure S5). It also provides stable reconstruction quality across all metrics and rendering variants (Figure S4).

S3.1.2 Runtime profile. Figure S7 provides a runtime breakdown of our Mode 3 tile-based gather accumulation under two scene complexities and varying holographic depth volumes. The holographic

depth volume controls the maximum subhologram footprint on the SLM: a larger depth range leads to larger subhologram supports and therefore more tile–source pairs to be processed. As the depth volume increases from 3 mm to 18 mm, the maximum subhologram size grows accordingly, resulting in a gradual increase in total rendering time for both 500K- and 1M-primitive scenes. The dominant cost comes from CUDA radix sorting, which organizes tile–source pairs for gather-based accumulation, while the costs of hit-pair generation and rendering remain relatively small. This indicates that the proposed gather formulation keeps the actual hologram accumulation efficient even under larger subhologram footprints, with the main scaling bottleneck shifted to sorting and data organization.

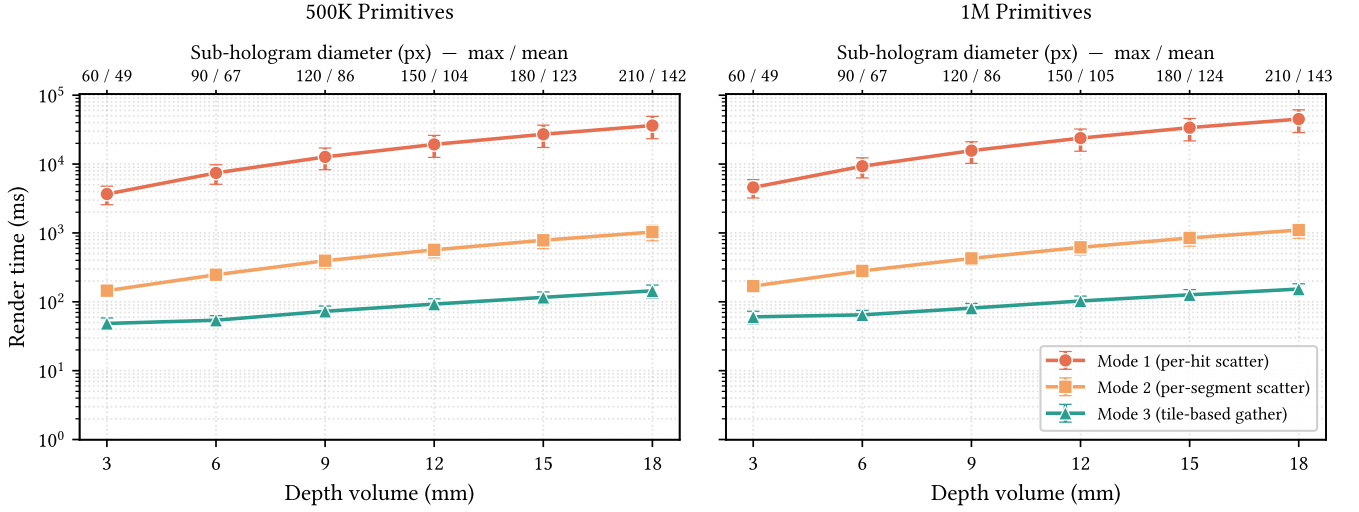


Fig. S6. Render time as a function of the holographic depth volume (with the corresponding sub-hologram radius shown on the top axis), evaluated at primitive caps of 500K and 1M. Each panel compares the three holographic ray-tracing variants – Mode 1 (per-hit scatter), Mode 2 (per-segment scatter), and Mode 3 (tile-based gather). Mode 1 and Mode 2 grow steeply with the depth volume because their per-ray wave-accumulation cost scales with the sub-hologram footprint, whereas Mode 3 remains nearly flat thanks to its tile-based gather formulation.

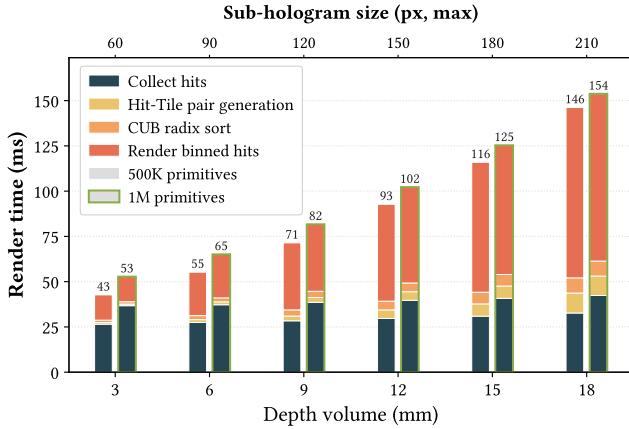


Fig. S7. Per-stage runtime breakdown of Mode 3 at 1024×1024 resolution as the holographic depth volume increases. For each depth setting, stacked bars report runtimes for 500K and 1M primitives, with the latter outlined in green. Each bar decomposes the total runtime into hit collection, hit–tile pair generation, CUB radix sort, and tile-based gather overheads; numbers above bars show total runtime in milliseconds. The top axis reports the maximum per-point sub-hologram size, which is determined by the depth volume. The breakdown shows that binning, sorting, and gather overheads remain modest as depth volume and primitive count increase, with the runtime dominated mainly by hit collection.

S3.2 Additional Visual Results

More captured results for our method on hybrid scenes are presented in Figure S8–S9. We also provide additional simulation results in Figure S10–Figure S14, in which we provide simulated reconstructions

from different methods’ generated holograms on various single-primitive scenes under multiple views. In Figure S15–S18, we provide additional captured results.

S3.3 Discussion of Key Modeling Choices

S3.3.1 Segment-wise virtual-source aggregation. As introduced in Eq. (9) of the main paper, WaveCast aggregates the volumetric interactions within each ray segment S_j into a single virtual source. Its depth and channel-wise amplitude are respectively given by the transport-weighted mean and weighted aggregation. For convenience, we restate the formulation here:

$$\bar{z}^{(j)} = \frac{\sum_{i \in S_j} w_i z_i}{\sum_{i \in S_j} w_i}, \quad \bar{a}_c^{(j)} = \sum_{i \in S_j} w_i c_{i,c}(\mathbf{d}), \quad (\text{S25})$$

where w_i is the transmittance-aware weight defined in Eq. (2) of the main paper. This transport-weighted construction is consistent with alpha-composited depth formulations used in prior Gaussian-splatting methods, where the visibility weights employed for RGB rendering are also used to obtain a representative ray depth [Huang et al. 2024; Matsuki et al. 2024; Turkulainen et al. 2025].

A natural question is whether a ray segment may contain multiple spatially separated dominant contributions, in which case the transport-weighted mean depth could lie between them rather than near either one. We examine this configuration on seven Mip-NeRF360 pure-GS scenes. For each scene, we sample 5,000 rays from the training views using each of two random seeds, yielding $7 \times 2 \times 5,000 = 70,000$ rays in total. For each ray, we first uniformly select a training view, uniformly sample a pixel within that view, and construct the corresponding ray using the camera intrinsics and pose. We retain rays with sufficient accumulated Gaussian contribution, $\sum_i w_i \geq 0.1$, resulting in 69,941 retained rays.

As summarized in Table S1, 12,233 retained rays (17.49%) contain multiple spatially separated dominant contribution clusters. The presence of multiple clusters alone does not necessarily make the representative depth ambiguous, since their transport weights may be unbalanced and the weighted mean may remain close to the stronger cluster. Among these multi-cluster cases, only 3,875 rays place the transport-weighted mean in the low-contribution region between the clusters, accounting for 5.54% of all retained rays.

Within this 5.54% subset, the transport-weighted mean depth deviates from the nearest dominant opacity peak by a pooled median of approximately $56 \mu\text{m}$ after mapping to the 3–8 mm holographic depth range. A small tail of strongly layered segments may produce larger depth errors. For such cases, a straightforward remedy is to adaptively subdivide the segment according to its weighted depth variance and generate multiple virtual sources, providing a controllable accuracy–efficiency trade-off.

S3.3.2 Effective image-depth modeling. Reflected and refracted content should be focused at its apparent image depth rather than at the physical surface depth. WaveCast computes this depth by recursively applying the reflection and refraction transforms described in Section S1.4. To evaluate their accuracy, we compare the predicted image positions with those obtained from non-paraxial vector reflection and Snell ray tracing.

We perform this controlled analysis on the mirror and glass primitives from two representative hybrid scenes, comprising one indoor scene (Bonsai) and one outdoor scene (Garden). For each primitive, R denotes its curvature radius, d_{cam} the distance from the camera to the primitive center, and s_{obj} the axial object distance measured from the primitive surface. The tested primitives lie at $d_{\text{cam}} \in [2.4, 8.8]$ world units and occupy 28–69% of the camera field of view, providing substantial curved-surface footprints.

Across every tested mirror and glass configuration, the resulting depth error is below $24 \mu\text{m}$ in display space, more than an order of magnitude smaller than the axial distance over which our display can resolve a change in focus, about $400 \mu\text{m}$.² The largest error occurs for the Garden glass sphere at the closest sampled object distance (3 world units), where the steepest ray our display can reproduce still meets the sphere at only 29% of its radius from the axis; for the other three primitives the error falls below $\sim 6 \mu\text{m}$. Per-primitive results are reported in Table S2.

²We use the conservative coherent criterion $\lambda/\sin^2 \theta_{\text{max}}$, with θ_{max} the maximum diffraction angle of Equation (S6).

Table S1. Empirical analysis of segment-wise virtual-source depth over seven Mip-NeRF360 pure-GS scenes. Percentages in the last two rows are computed over the 69,941 retained rays; the final row is a subset of the preceding multi-cluster category.

Ray category	Count	Percentage
Sampled rays	70,000	100%
Retained rays ($\sum_i w_i \geq 0.1$)	69,941	99.92%
Two dominant clusters	12,233	17.49%
Mean depth between clusters	3,875	5.54%

Table S2. Controlled evaluation of the reflection and refraction transforms used for effective image depth. The relative image-depth error is $|s'_{\text{ray}} - s'_{\text{par}}|/|s'_{\text{par}}|$. The display-space error is obtained by applying the same per-scene depth mapping used by the hologram renderer. Each row reports the largest error over $s_{\text{obj}} \in \{3, 5, 10, 20\}$ world units.

Scene	Primitive	R (world)	Rel. depth error	Display error (μm)
Bonsai	Mirror	1.00	2.6%	1.5
Bonsai	Glass [†]	0.42	5.4%	2.5
Garden	Mirror	1.00	10.3%	5.8
Garden	Glass	1.00	13.2%	24.0

[†]The non-uniformly scaled Bonsai glass primitive is evaluated using an equivalent sphere with its mean transformed radius.

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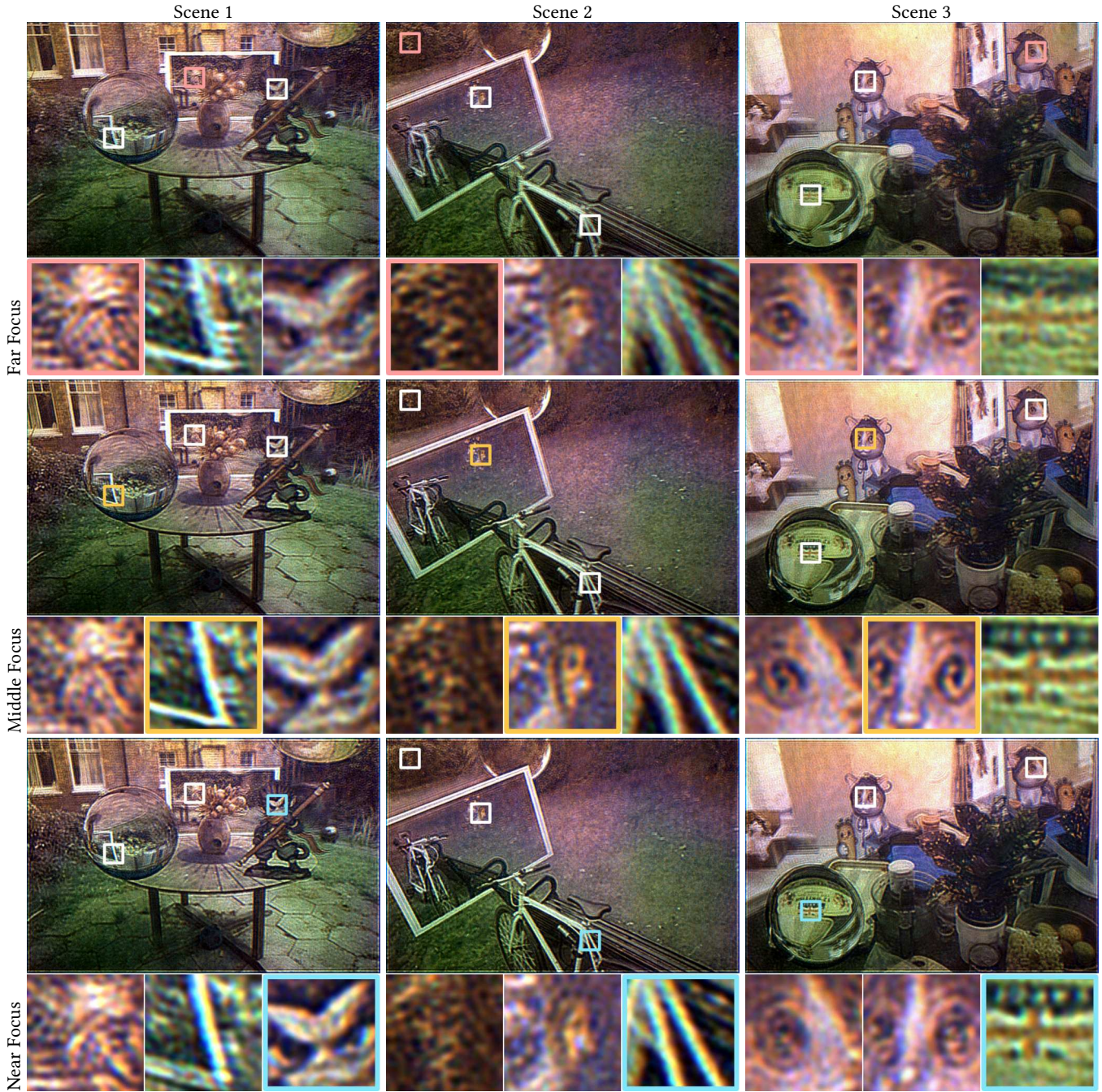


Fig. S8. **Experimentally captured reconstructions on hybrid scenes.** WaveCast is evaluated on three hybrid scenes that combine real 3DGS reconstructions with inserted mesh-based diffuse, reflective, and refractive objects. Captured focal reconstructions at far, middle, and near focus distances show refocusable real and virtual content, with zoom-in regions highlighting scene-dependent occlusion and optical interactions. Reflective and refractive objects alter not only local appearance but also the focus behavior of the observed content, demonstrating the benefit of carrying ray-path information into hologram synthesis.



Fig. S9. **Experimentally captured reconstructions on hybrid scenes.** WaveCast is evaluated on another three hybrid scenes that combine real 3DGS reconstructions with inserted mesh-based diffuse, reflective, and refractive objects. Captured focal reconstructions at far, middle, and near focus distances show refocusable real and virtual content, with zoom-in regions highlighting scene-dependent occlusion and optical interactions. Reflective and refractive objects alter not only local appearance but also the focus behavior of the observed content, demonstrating the benefit of carrying ray-path information into hologram synthesis.

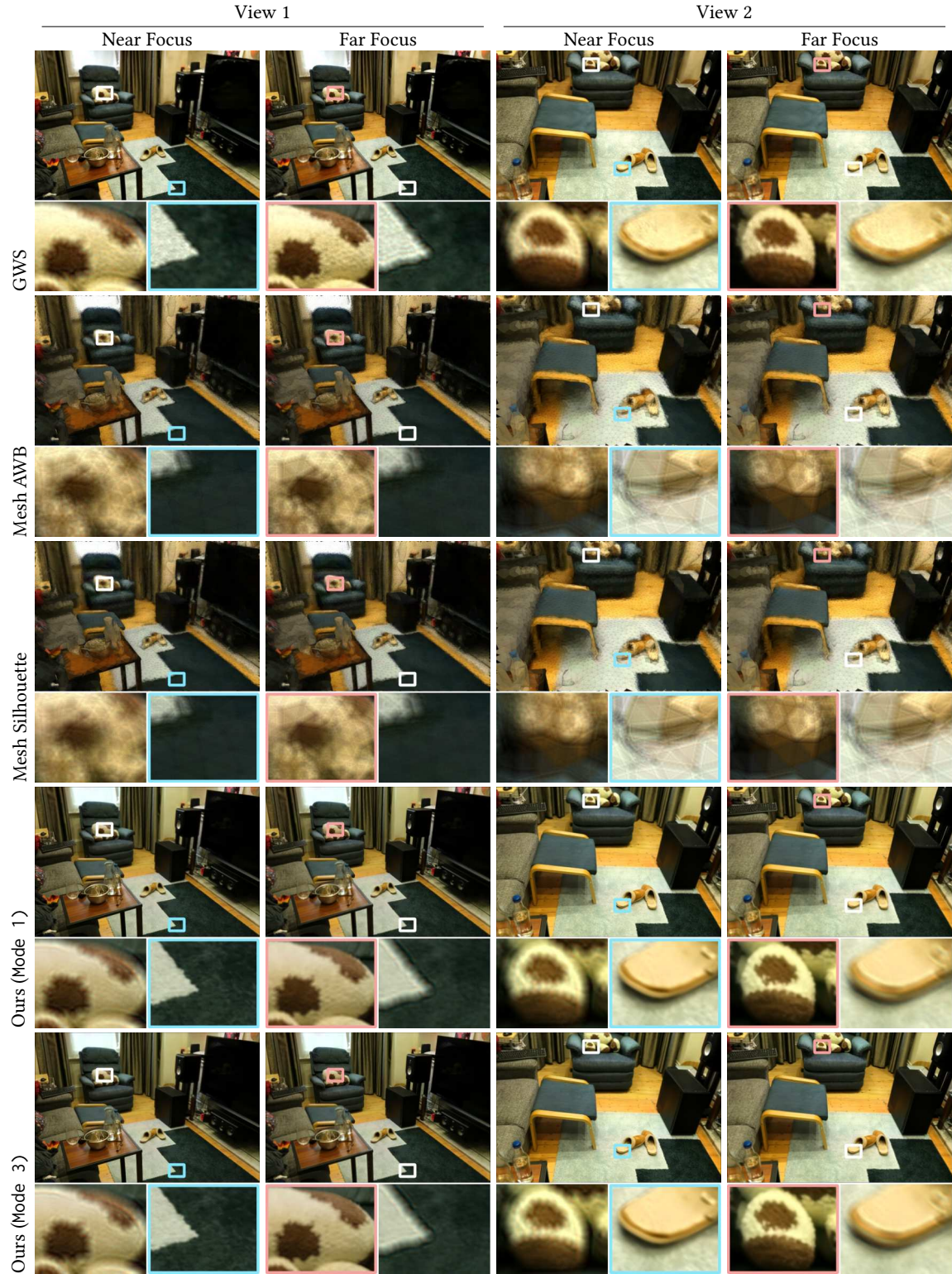


Fig. S10. Simulated reconstructions for holograms produced by different methods.

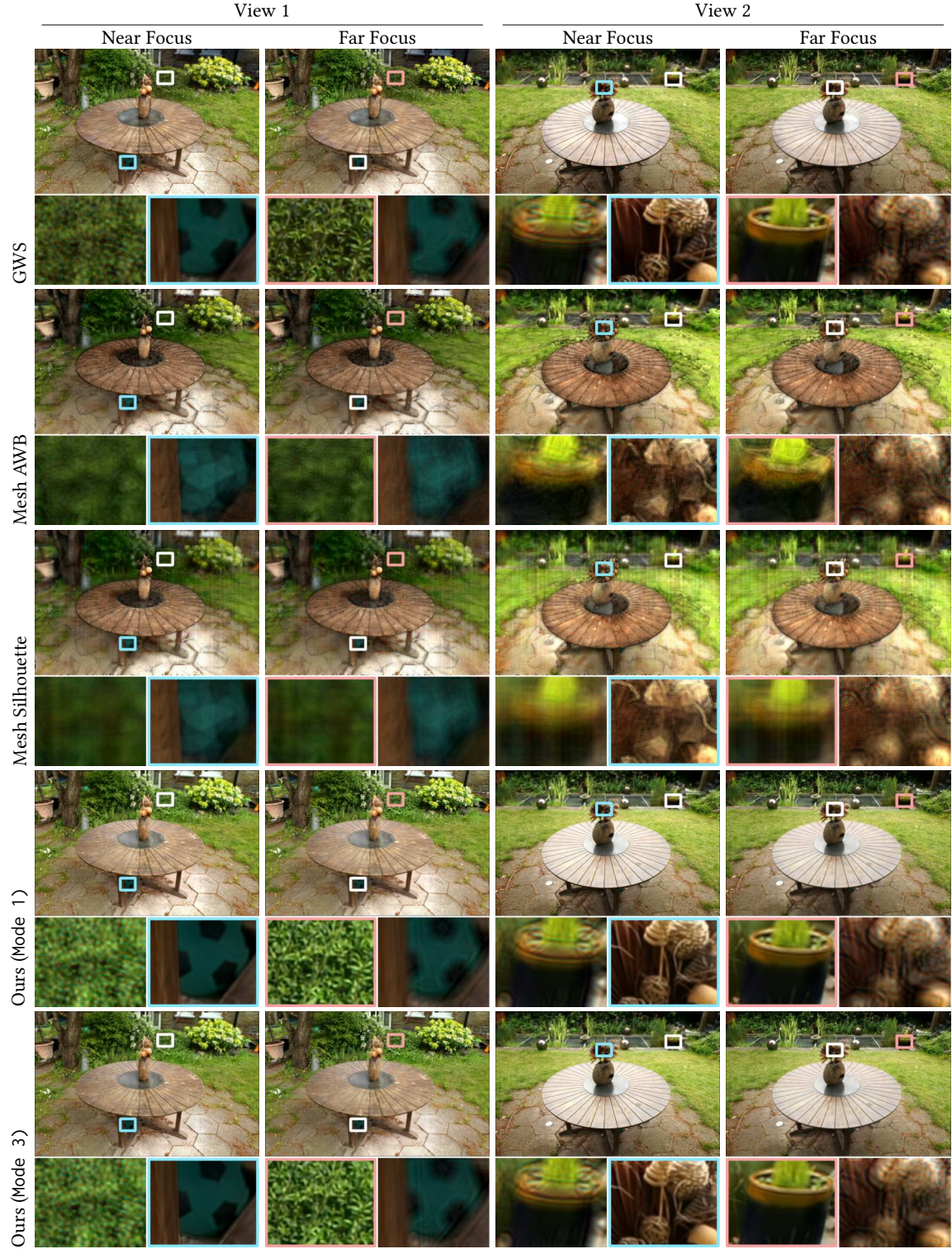


Fig. S11. Simulated reconstructions for holograms produced by different methods.

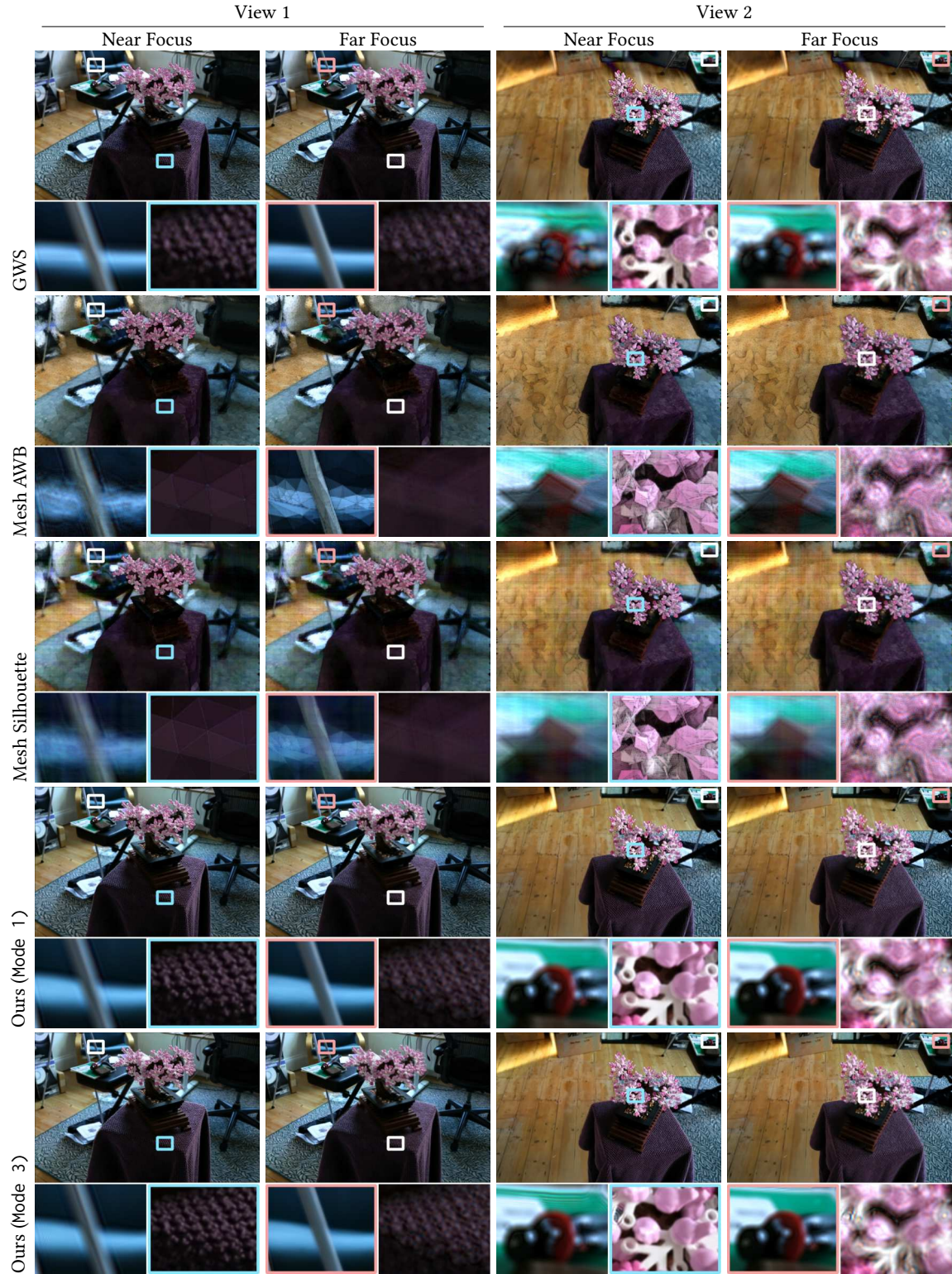


Fig. S12. Simulated reconstructions for holograms produced by different methods.

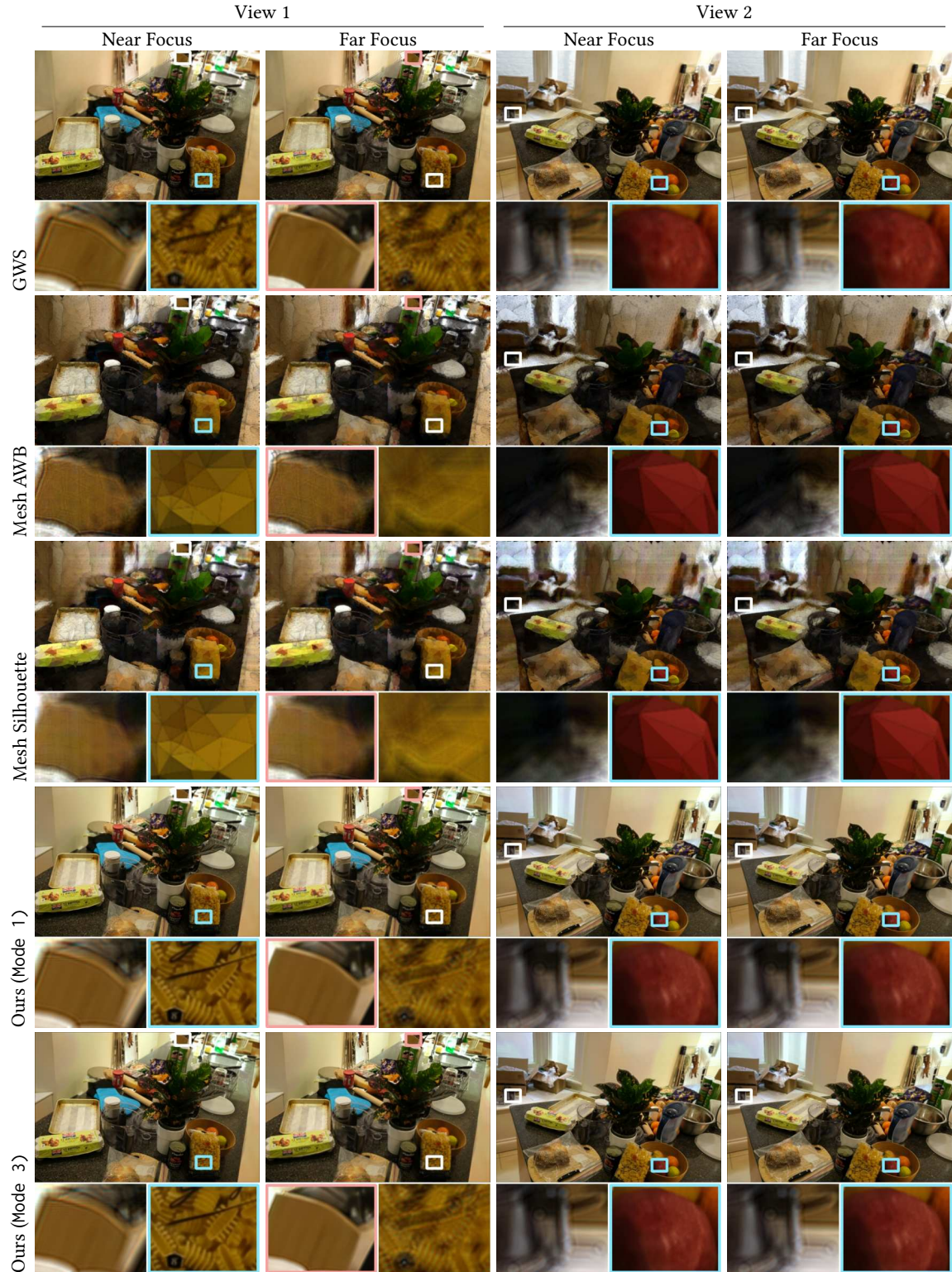


Fig. S13. Simulated reconstructions for holograms produced by different methods.

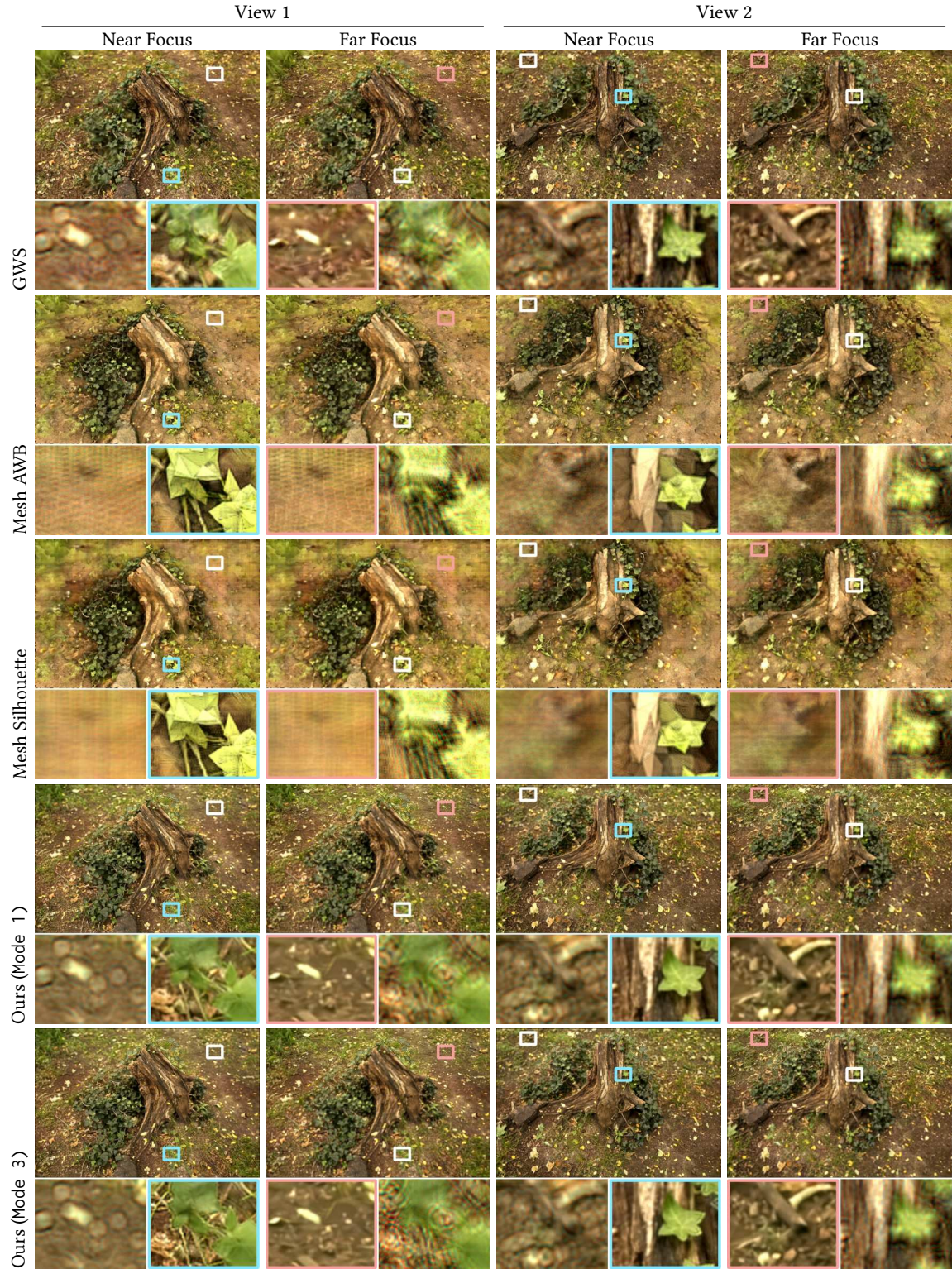


Fig. S14. Simulated reconstructions for holograms produced by different methods.

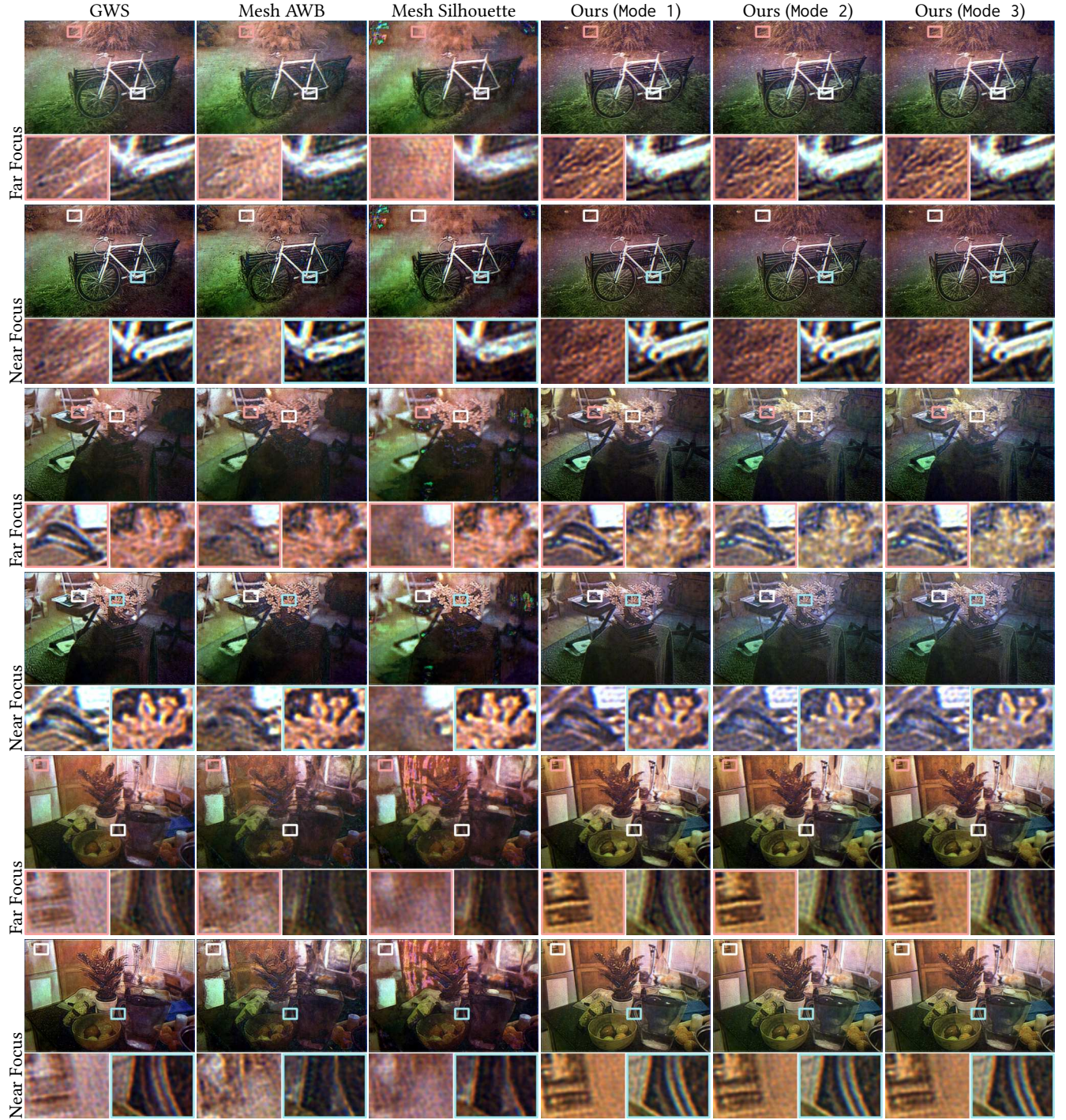


Fig. S15. Extended experimental results for different methods' holograms on various scenes.



Fig. S16. Extended experimental results for generated holograms from different methods under multiple views.

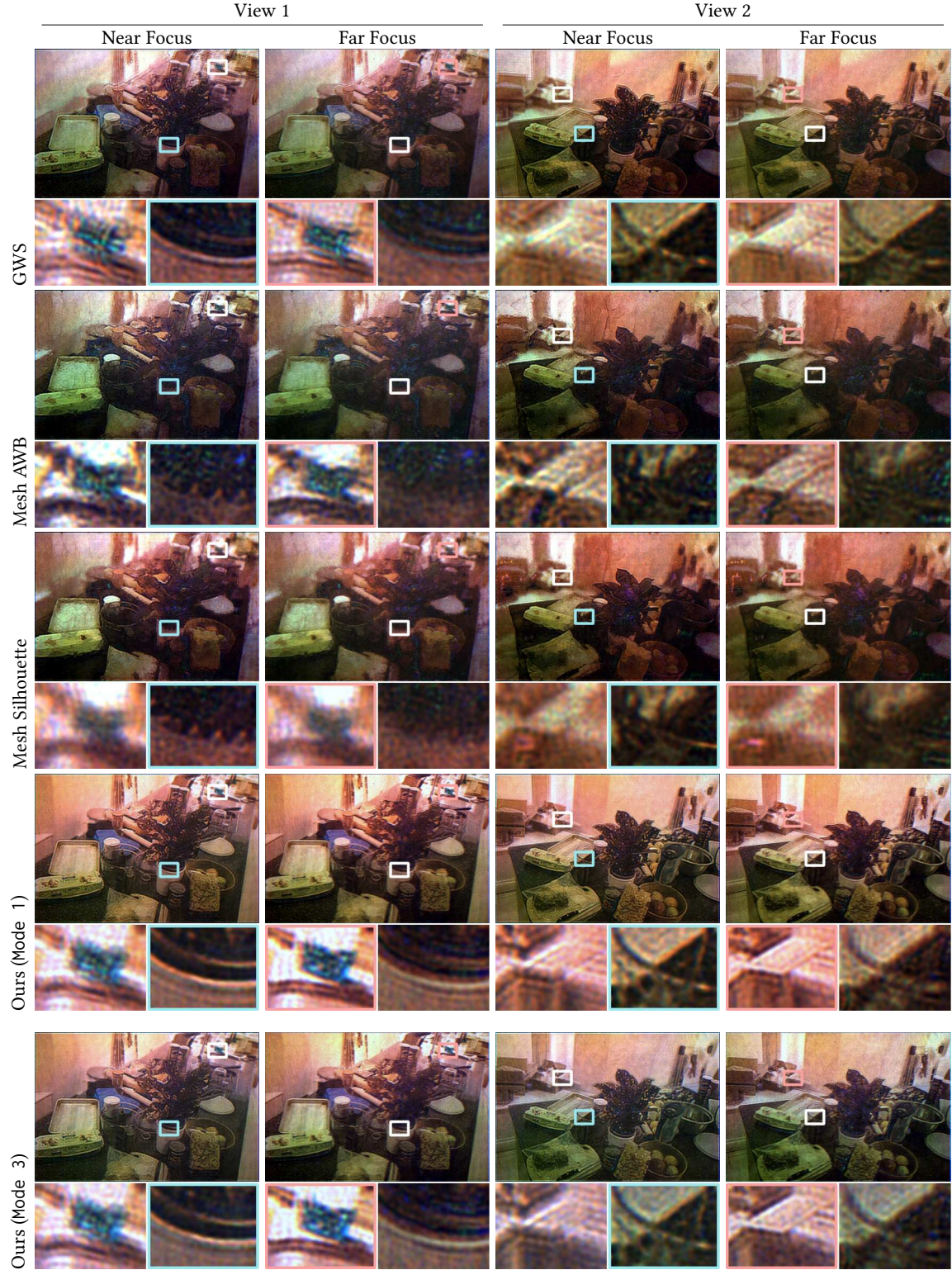


Fig. S17. Extended experimental results for generated holograms from different methods under multiple views.

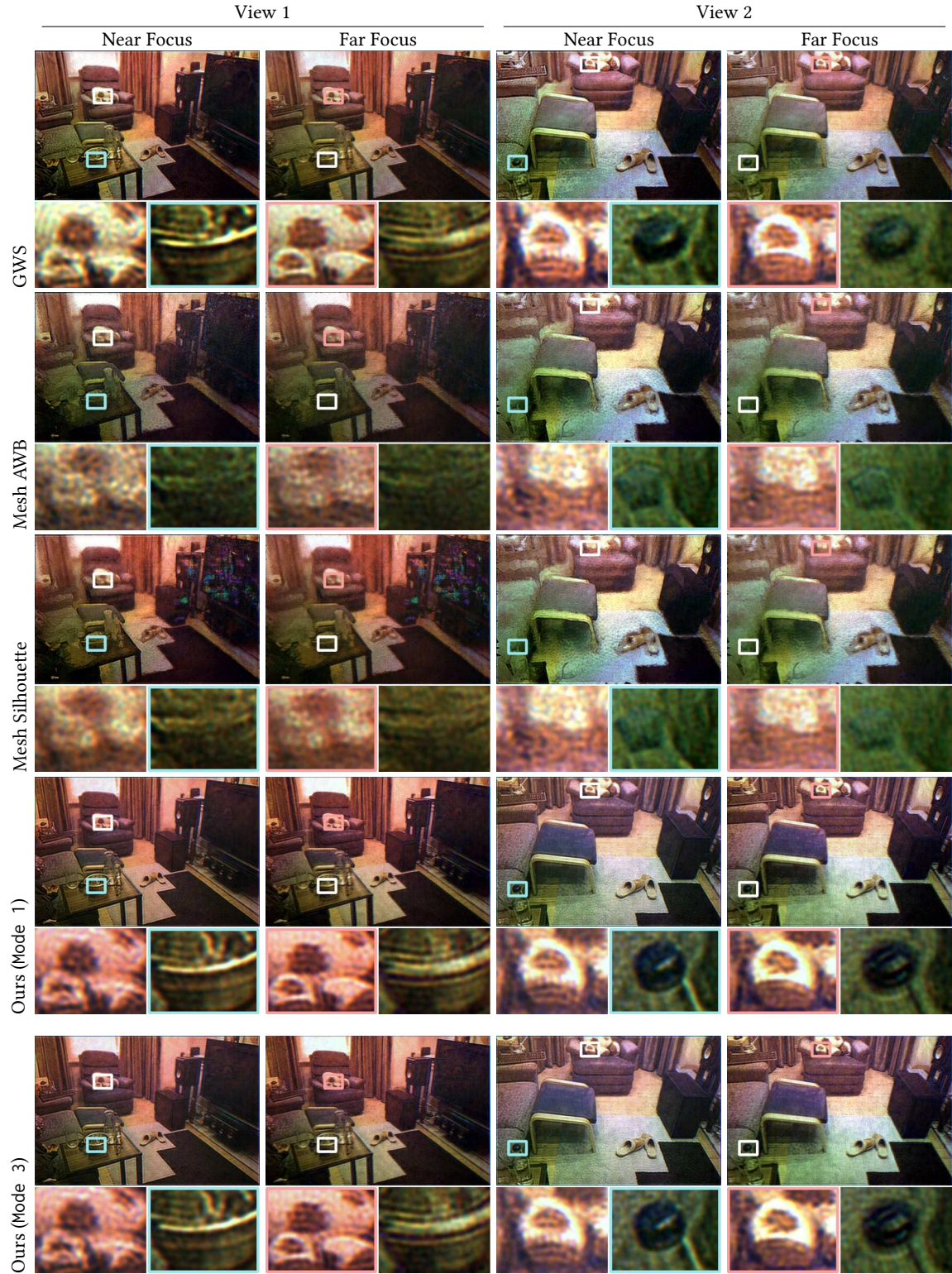


Fig. S18. Extended experimental results for generated holograms from different methods under multiple views.